

AI/ML intern DAY 14

Team Task: Cloth Simulation Research Objective

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1. Introduction

Cloth simulation is one of the most critical components of modern Virtual Try-On systems. It enables digital garments to behave realistically when placed on virtual avatars or human body models. The primary objective of cloth simulation is to reproduce natural fabric behaviour such as folding, stretching, wrinkling, collision response, and movement under different conditions.

In Virtual Try-On applications, cloth simulation significantly impacts user experience because it determines how accurately a garment fits and reacts to body movement. Realistic cloth behaviour improves immersion, increases customer confidence, and enhances the overall effectiveness of digital fashion platforms.

This research focuses on studying four widely used cloth simulation technologies:

- Unity Cloth System
- NVIDIA PhysX
- Marvelous Designer
- Blender Cloth Physics

The study evaluates each technology based on realism, performance, collision handling, ease of use, and suitability for AR/VR Virtual Try-On systems.

2. Research Objectives

The main objectives of this research are:

- Study modern cloth simulation technologies
 - Understand realistic garment behaviour
 - Analyse cloth collision handling techniques
 - Evaluate simulation performance
 - Compare industry-standard tools
 - Identify the most suitable technology for Virtual Try-On systems
 - Provide recommendations for future implementation
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3. Importance of Cloth Simulation in Virtual Try-On Systems

Virtual Try-On systems require garments to behave similarly to real-world clothing. Without proper cloth simulation, garments may appear unrealistic, float above the body, penetrate body surfaces, or fail to adapt to user movements.

Cloth simulation improves:

- Garment realism
- User experience
- Virtual fitting accuracy
- Fabric behaviour representation
- Customer confidence in online shopping
- AR/VR immersion

A successful cloth simulation system must support realistic movement while maintaining real-time performance.

4. Unity Cloth System

Overview

Unity Cloth System is a built-in cloth simulation framework available within the Unity game engine. It is commonly used in games, AR applications, VR environments, and interactive simulations.

The system allows developers to simulate clothing behaviour directly on animated characters using real-time physics calculations.

Key Features

- Real-time cloth simulation
- Cloth collision detection
- Wind effects
- Character clothing support
- Physics-based garment behaviour
- Easy integration with avatars

Advantages

- Easy implementation
- Suitable for AR and VR applications
- Good real-time performance
- Built-in Unity support

- Strong community support

Limitations

- Less realistic than specialized fashion software
- Limited garment design capabilities
- Complex garments may require optimization

Applications

- Virtual Try-On systems
- Gaming environments
- AR fashion applications
- VR simulations

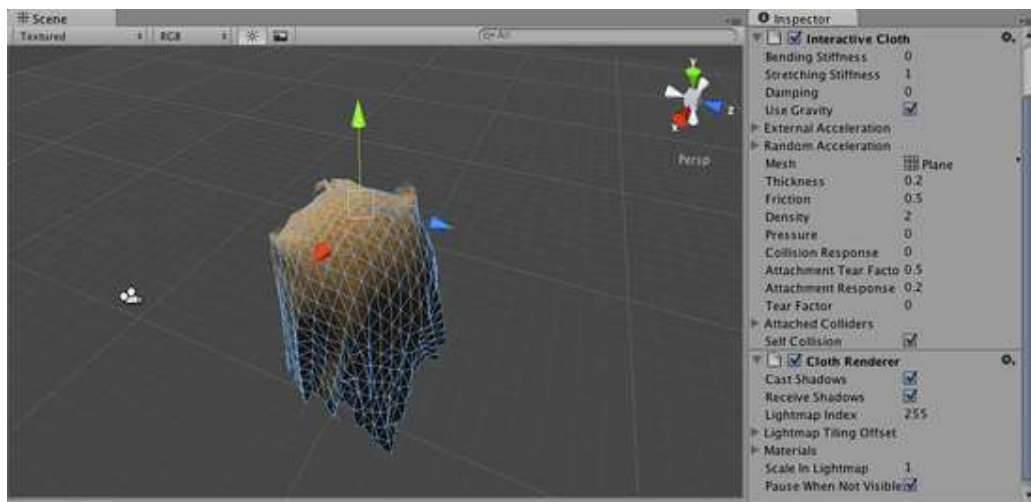


Figure 1: Unity Cloth System demonstrating real-time cloth simulation on a 3D character model.

5. NVIDIA PhysX

Overview

NVIDIA PhysX is an advanced physics engine designed to simulate real-world physical interactions. It provides highly optimized cloth simulation capabilities and is widely used in gaming, virtual production, and interactive 3D applications.

PhysX focuses on realistic cloth movement while maintaining high computational efficiency.

Key Features

- Advanced cloth simulation
- GPU acceleration
- Collision handling
- Physics-based deformation

- Real-time performance
- High scalability

Advantages

- Industry-standard technology
- Excellent collision detection
- High-performance simulation
- GPU optimization support
- Suitable for large-scale projects

Limitations

- Requires powerful hardware for complex simulations
- More difficult to configure than basic solutions
- Advanced optimization may be necessary

Applications

- AAA games
- AR/VR applications
- Virtual fashion systems
- Real-time simulation environments

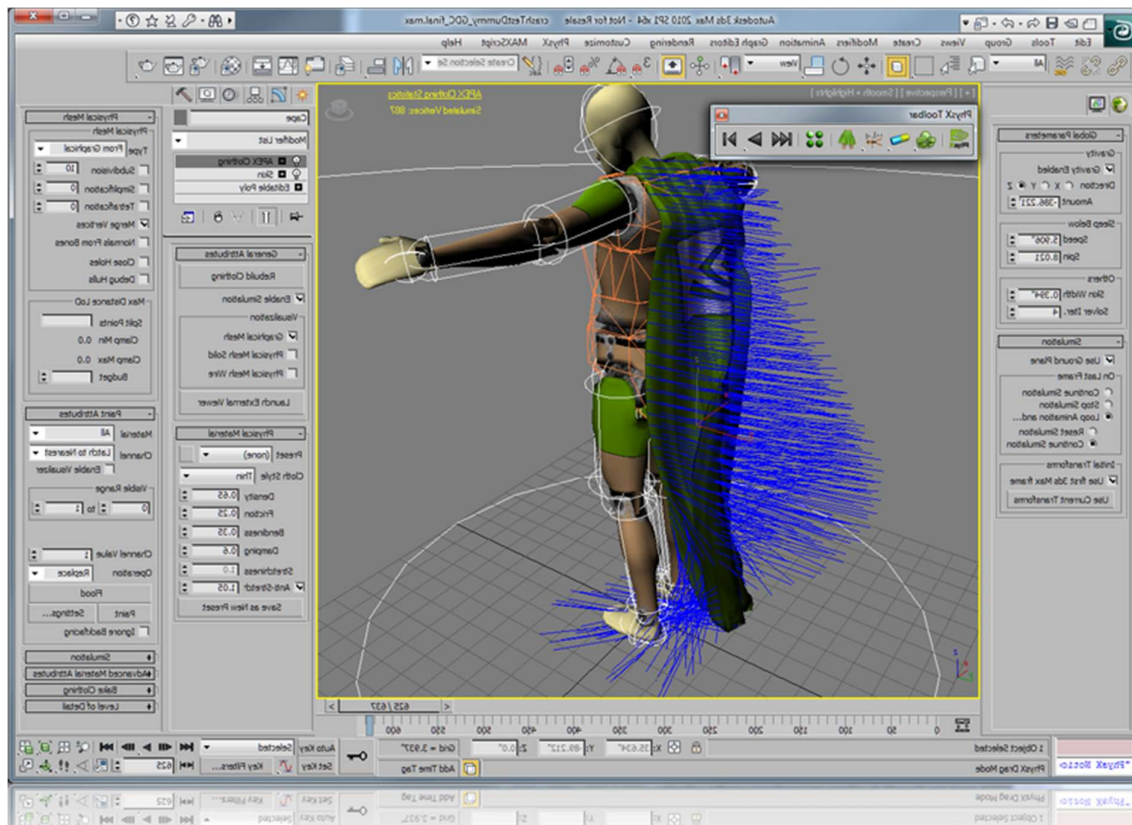


Figure 2: NVIDIA PhysX cloth simulation showing advanced collision handling and realistic fabric movement.

6. Marvelous Designer

Overview

Marvelous Designer is a professional garment design and cloth simulation software widely used in the fashion, gaming, and animation industries.

Unlike traditional physics engines, Marvelous Designer uses pattern-based garment construction similar to real-world clothing manufacturing processes.

Key Features

- Pattern-based garment creation
- Fabric property customization
- Realistic wrinkle generation
- Advanced cloth simulation
- Garment stitching tools
- High-quality visualization

Advantages

- Highest level of realism
- Fashion industry standard
- Excellent cloth behaviour simulation
- Accurate garment construction
- Superior fabric representation

Limitations

- Commercial software
- High hardware requirements
- Not designed for real-time applications

Applications

- Fashion design
- Digital clothing production
- Film production
- Character costume design
- Virtual garment creation



Figure 3: Marvelous Designer demonstrating realistic garment simulation, fabric deformation, and cloth behaviour on an animated character model.

7. Blender Cloth Physics

Overview

Blender Cloth Physics is the built-in cloth simulation system available within Blender. It provides realistic cloth behaviour while remaining completely free and open source.

Blender is widely used by students, researchers, independent developers, and production studios.

Key Features

- Cloth simulation
- Collision detection
- Wind simulation
- Fabric parameter customization
- Garment animation
- Open-source platform

Advantages

- Completely free
- Strong community support
- Good simulation quality
- Flexible workflow

- Easy integration with 3D assets

Limitations

- Not optimized for real-time AR applications
- Longer simulation times
- Requires manual optimization

Applications

- Educational projects
- Virtual Try-On research
- Animation production
- Fashion visualization
- Digital garment testing

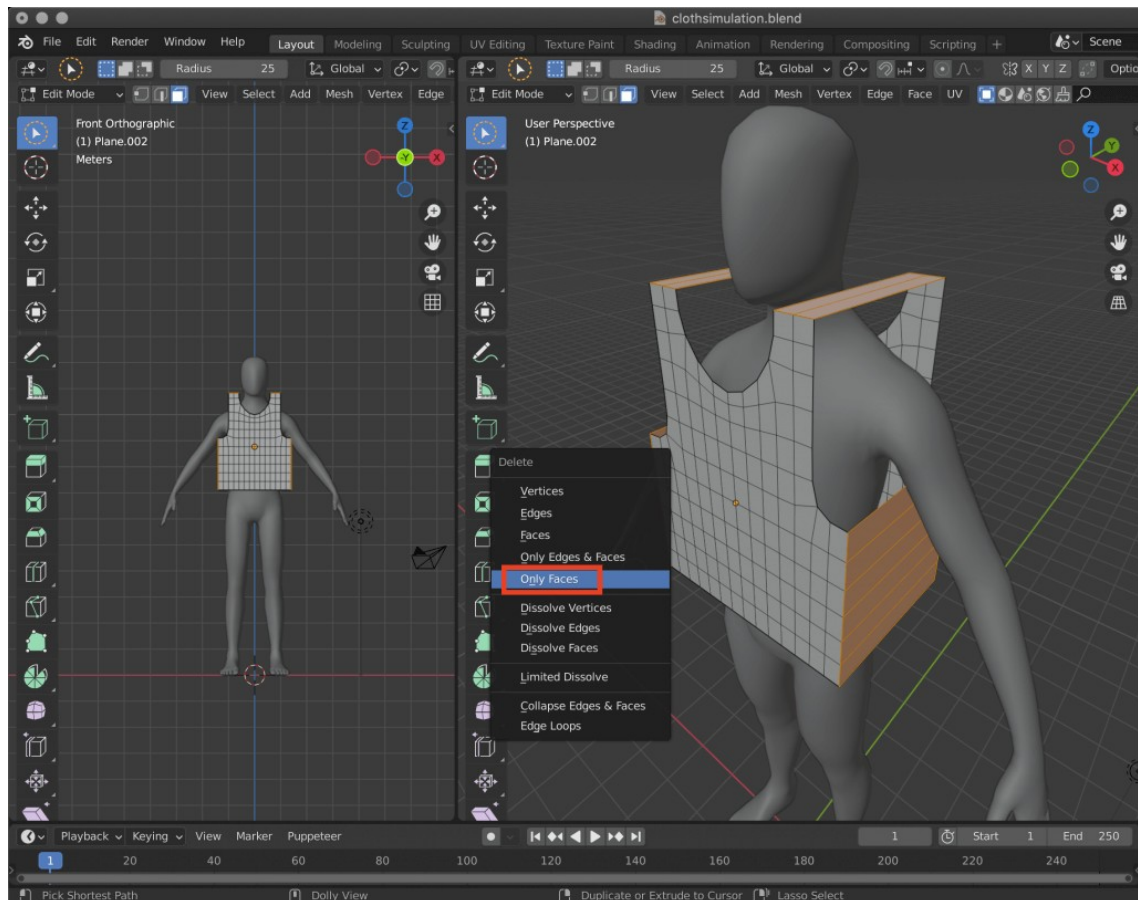


Figure 4: Blender Cloth Physics interface showing collision configuration and cloth simulation settings used for garment interaction with 3D character models.

8. Feature Comparison Table

Feature	Unity Cloth	NVIDIA PhysX	Marvelous Designer	Blender Cloth
Real-Time Simulation	Yes	Yes	No	Limited
Cloth Realism	Medium	High	Very High	High
Collision Handling	Good	Excellent	Excellent	Good
AR/VR Compatibility	Excellent	Excellent	Limited	Limited
Ease of Use	Easy	Medium	Medium	Easy
Cost	Free	Free	Paid	Free
Industry Adoption	High	High	Very High	High
Learning Curve	Low	Medium	Medium	Low
Performance	High	Very High	Medium	Medium
Research Suitability	Good	Excellent	Excellent	Excellent

9. Performance Analysis

Cloth Movement

Marvelous Designer provides the most realistic cloth movement due to its pattern-based garment simulation system.

Best Technology

Marvelous Designer

Reason:

- Natural garment behaviour
- Realistic folds
- Accurate wrinkle formation

Collision Handling

NVIDIA PhysX demonstrates superior collision handling capabilities.

Best Technology

NVIDIA PhysX

Reason:

- Advanced physics calculations

- Strong body-cloth interaction
- Excellent collision accuracy

Fabric Realism

Marvelous Designer provides the highest fabric realism.

Best Technology

Marvelous Designer

Reason:

- Fashion industry workflow
- Advanced fabric simulation
- Accurate material behaviour

Real-Time Performance

Unity Cloth System combined with NVIDIA PhysX offers the best real-time performance.

Best Technology

Unity + PhysX

Reason:

- Optimized for real-time applications
- Suitable for AR/VR environments
- Fast simulation speed

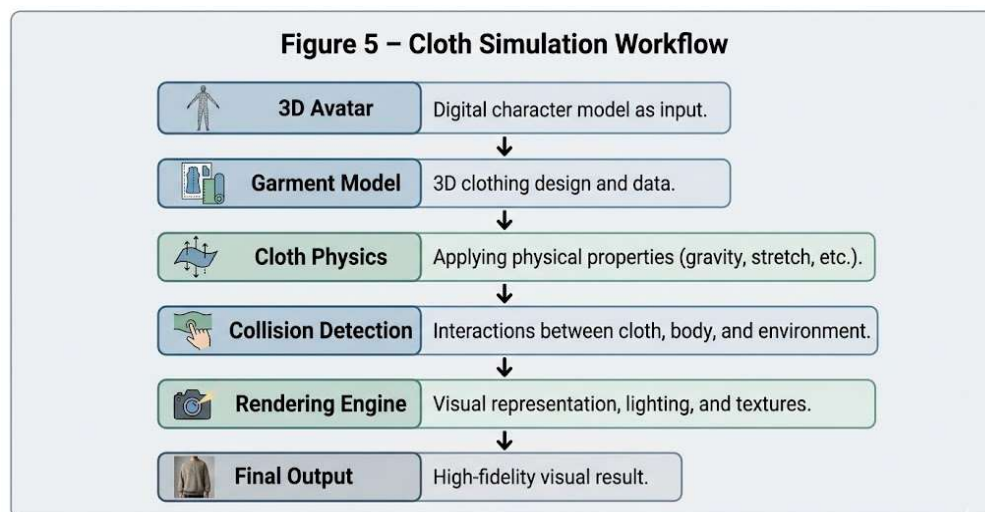


Figure 5: General workflow of cloth simulation in Virtual Try-On systems.

10. Industry Usage Analysis

Gaming Industry

Most commonly used:

- Unity Cloth System
- NVIDIA PhysX

Reason:

- Real-time rendering requirements
- High performance demands

Fashion Industry

Most commonly used:

- Marvelous Designer

Reason:

- Realistic garment creation
- Professional fashion workflows

Research and Education

Most commonly used:

- Blender Cloth Physics

Reason:

- Free availability
 - Open-source ecosystem
 - Strong academic adoption
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11. Challenges Identified

Several challenges were observed during the research process.

Performance Challenges

- High computational requirements
- GPU dependency
- Large simulation workloads

Collision Challenges

- Cloth penetration
- Body intersection
- Collision accuracy

Realism Challenges

- Fabric representation
- Wrinkle generation
- Dynamic garment behaviour

Scalability Challenges

- Real-time constraints
 - Multi-user environments
 - Mobile device limitations
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12. Recommended Solution for Virtual Try-On Systems

Based on the research findings, the recommended implementation strategy is:

Blender + Unity + NVIDIA PhysX

Reasoning

Blender:

- Create garments
- Prepare assets
- Prototype clothing

Unity:

- Real-time rendering
- AR/VR integration
- User interaction

NVIDIA PhysX:

- Cloth simulation
- Collision handling
- Physics calculations

This combination provides:

- High realism
- Strong performance
- AR/VR compatibility
- Scalability
- Cost effectiveness



Figure 6: Example of a Virtual Try-On system using cloth simulation technologies.

13. Future Improvements

Potential future enhancements include:

- AI-based cloth simulation
- Neural garment deformation
- Real-time physics optimization
- Personalized body fitting
- Advanced fabric material modelling
- Cloud-based simulation systems
- GPU acceleration improvements

14. Conclusion

This research analysed four major cloth simulation technologies used in modern Virtual Try-On systems: Unity Cloth System, NVIDIA PhysX, Marvelous Designer, and Blender Cloth Physics.

The study found that each technology offers unique strengths depending on project requirements. Marvelous Designer provides the highest realism, NVIDIA PhysX delivers the best collision handling, Unity Cloth System offers strong real-time performance, and Blender Cloth Physics provides an accessible and cost-effective research platform.

For Virtual Try-On applications requiring a balance between realism and performance, the combination of Blender, Unity, and NVIDIA PhysX is recommended as the most practical and scalable solution.

The research establishes a strong foundation for future implementation of realistic digital garment fitting systems in AR, VR, and AI-powered fashion technology platforms.